

MUN, Jihyun

Postdoctoral Researcher · Samovar, Télécom SudParis, Institut Polytechnique de Paris

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RESEARCH INTERESTS

Atypical / Disordered speech analysis

Spoken language processing technology

Natural language processing technology

Acoustic analysis, Phonetic analysis, Phonemic analysis, Linguistic analysis

Online handwriting analysis, Developmental dysgraphia

Clinically reliable and interpretable machine learning for health

EDUCATION

Integrated M.A./Ph.D., Linguistics, Seoul National University	Mar 2021 – Aug 2025
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B.A., Mechanical Engineering, B.A., Linguistics, Seoul National University	Mar 2015 – Feb 2021
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Anyang Foreign Language High School (Chinese major)	Mar 2012 – Feb 2015
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POSITIONS

Postdoctoral Researcher, Télécom SudParis	Jan 2026 – Present
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PUBLICATIONS

CONFERENCES

Mun, J., Kim, S., & Chung, M. (2025). Speech-Based Automatic Chronic Kidney Disease Diagnosis via Transformer Fusion of Glottal and Spectrogram Features. Proc. Interspeech 2025.

Mun, J., Kim, S., & Chung, M. (2025). A Cascaded Multimodal Framework for Automatic Social Communication Severity Assessment in Children with Autism Spectrum Disorder. Proc. Interspeech 2025.

Mun, J., Kim, S., Chung, M. (2024). Developing an End-to-End Framework for Predicting the Social Communication Severity Scores of Children with Autism Spectrum Disorder. Accepted to Interspeech 2024.

Lee, S., **Mun, J.**, Kim, S., Park, H., Yang, S., Kim, H., Noh, S., Kim, W., & Chung, M. (2024). Automatic Speech Recognition and Assessment Systems Incorporated into Digital Therapeutics for Children with Autism Spectrum Disorder. Accepted to ICCHP 2024.

Lee, S., **Mun, J.**, Kim, S., & Chung, M. (2024). Speech Corpus for Korean Children with Autism Spectrum Disorder: Towards Automatic Assessment Systems. Accepted to LREC-Coling 2024.

Mun, J., Kim, S., Kim, M. J., Ryu, J., Kim, S., & Chung, M. (2023). An Analysis of Glottal Features of Chronic Kidney Disease Speech and Its Application to CKD Detection. Proc. Interspeech 2023.

Mun, J., Kim, S., Kim, M. J., Ryu, J., Kim, S., & Chung, M. (2022). A speech corpus for chronic kidney disease. Proc. Oriental COCOSDA 2022.

JOURNALS

Mun, J., Kim, S., Kim, M. J., Ryu, J., Kim, S., & Chung, M. (2022). Automatic detection and severity prediction of chronic kidney disease using machine learning classifiers. *Phonetics and Speech Sciences*, 14(4), 45-56.

Mun, J., Kim, S., & Chung, M. (2021). Acoustic analysis of Korean affricates produced by dysarthric speakers with cerebral palsy. *Phonetics and Speech Sciences*, 13(2), 45-55.

UNDER REVIEW

Mun, J., & El-Yacoubi, M. A. Profiling Handwriting-Process Deviations in Developmental Dysgraphia: An Open, Normatively-Referenced Instrument. Under review, *Behavior Research Methods*. Preprint: arXiv:2609.15435.

Mun, J., & El-Yacoubi, M. A. An Explainable Machine Learning Pipeline for Online-Handwriting-Based Dysgraphia Detection: Unbiased Evaluation, Feature Attribution, and Exploratory Subgroup Characterization. Under review, *IEEE Transactions on Cognitive and Developmental Systems*.

IN PREPARATION

Mun, J., & El-Yacoubi, M. A. Decomposing reported performance in small-sample clinical ML.

PROJECTS

Development of a Personalized Integrated System Combining Wearable Transcranial Magnetic Stimulation and Digital Therapeutics for Enhancing Communication in Children with Autism Spectrum Disorder	2025.04- 2025.12
Funded by the Korean Ministry of Trade, Industry, and Resources.	
Developed an automatic speech recognition system and automatic social communication severity classification system for children with autism.	
Development of Digital Therapeutics to Improve Communication Skills of Autistic Patients	2022.09- 2024.12
Funded by the Korean Institute for Information & Communication Technology Planning & Evaluation.	
Developed an automatic speech recognition system and automatic social communication severity classification system for children with autism.	
Construction of a Cohort through Voice Banking of Chronic Kidney Disease Patients and Analysis of Voice Characteristics according to Renal Function	2022.03- 2024.02
Funded by the Seoul National University Bundang Hospital.	
Designed a speech corpus of chronic kidney disease patients, analyzed speech characteristics of chronic kidney disease patients, and developed an automatic detection and severity classification system for chronic kidney disease patients.	
Acquiring and Utilizing Audio Big Data for Healthcare Service Design	2021.03- 2023.12
Funded by the Korean Ministry of Science and ICT.	
Designed a speech corpus of chronic kidney disease patients, analyzed speech characteristics of chronic kidney disease patients, and developed an automatic detection and severity classification system for chronic kidney disease patients.	
Korean Speech Data for Western and Asian Language Users for Language Education	2022.09- 2022.11
Funded by the Korean National Information Society Agency.	
Developed an automatic phonetic transcription toolkit.	
Multilingual Speech Data Collection for L2 Korean Learners	2022.07- 2022.11
Funded by the Korean National Information Society Agency.	
Developed an automatic phonetic transcription toolkit for L2 Korean speech.	
Development of Intelligent Tool-based Content Production and Enjoyment Support Technology Considering the Accessibility of the Weak in Social Communication	2021.03- 2021.12
Funded by the Korean National Information Society Agency.	
Developed an automatic speech recognition system for children with autism.	

PATENTS

Automated System for Predicting Social Communication Severity Scores of Children with Autism Spectrum Disorder and Method for Predicting Scores Using The Same
(Application number: 10-2024-0117393)

SOFTWARE REGISTRATIONS

Automatic Speech Recognition System for Semi-spontaneous Speech of Children with Autism Spectrum Disorder
(Korea Copyright Commission, C-2026-046494; registered 2026.09.17; rights held by SNU R&DB Foundation)

Social Communication Severity Assessment Model for Children with Autism
(Korea Copyright Commission, C-2024-033498; registered 2024.09.25; rights held by SNU R&DB Foundation)

GRANTS

Subsequent generations of basic studies, Seoul National University	Spring – Fall 2024
Travel Grant, College of Humanities, Seoul National University	LREC-Coling 2024
Travel Grant, College of Humanities, Seoul National University	Interspeech 2023
Subsequent generations of basic studies, Seoul National University	Spring – Fall 2023
Son Joo-eun, Creative Talent Scholarship Student, Seoul National University	Fall 2022

INTERNSHIP

2019 Summer Internship in R&D Department, College of Engineering, Seoul National University & Hyundai Motors Group

LANGUAGES

Korean (native), English (Fluent), French (A2), Chinese (reading), German (reading), Japanese (reading)